



(Don't Write
anything in this Area)

- 7.c) Discuss how ecological stresses influence both genetic and non-genetic human adaptations. Illustrate with examples from hot deserts and cold climates. (15 Marks)

Ecological stress are the environmental stress to the human body which make humans to adapt and acclimatise both genetically and culturally.

In the HOT deserts

The adjustment of the human in Genetic features are:-

- ADH gene → to reduce dehydration
- TSP27 gene → to relieve from excess heat
- Allen's rule & Bergman's rule.

Non-Genetic adaptation

Physiological changes

- More sweating → More Cardiac Output to release the heat

→ Vasodilation to vent out heat.

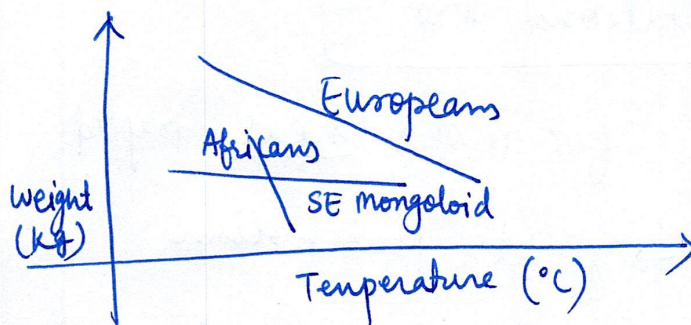


Fig: Berg's man rule.

In hot Condition

→ Body size is less to release heat more easily.

In cold: it is opposite

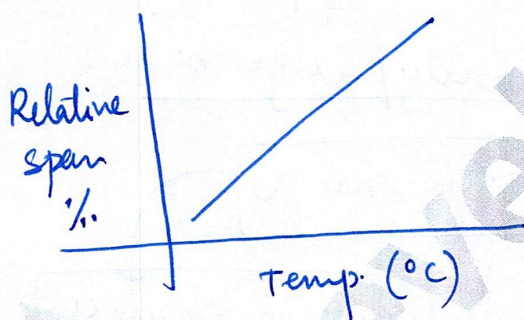


Fig: ~~Berg's man rule~~ Allen's rule

In hot: the lower legs & appendages are long

In cold: These are short to Conserve heat.

Cultural adaptation in Hot

→ Light clothes → half clothes.

→ More fruits, watermelon.

→ Use of ACs, Cooler or Cold water.

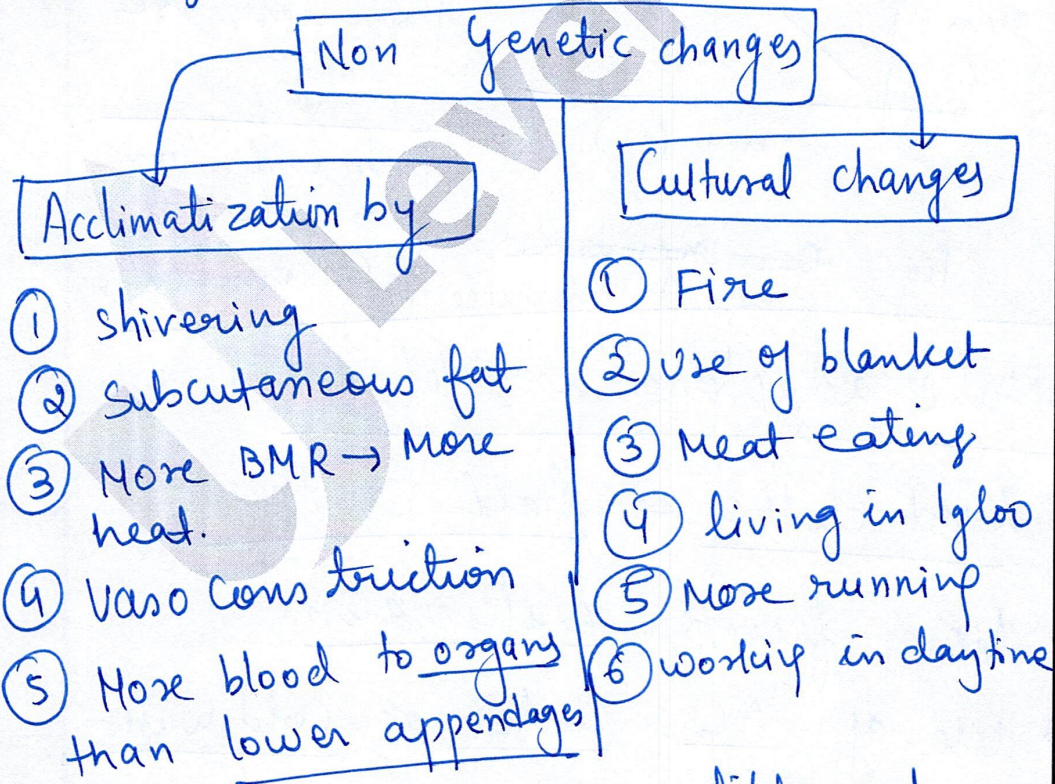
→ heat reflecting materials used for making home.

(eg) Nilotics of Africa; Pygmies.

In the cold climate:-

Genetic adaptations are:-

- ① UCP1 gene for more heat trapped
- ② FADS gene → greater metabolism
- ③ ADRA24 gene → for vaso constriction
- ④ Allen's rule: shorter appendages
- ⑤ Bergsmann rule: Fatty/ larger body.



Thus, we can see the different ways in which body adapts to its ecological stressors.